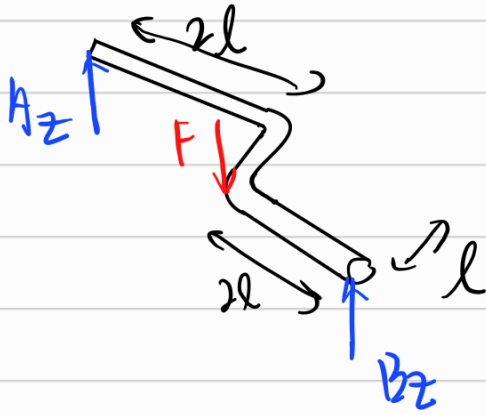
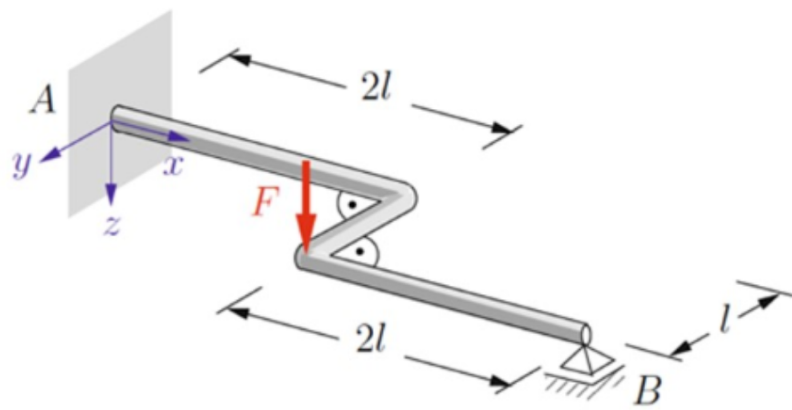
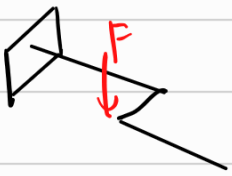


Homework 5.1 A steel frame ($E/G=3/8$) consists of three solid circular beam (radius r). Calculate the support reactions due to an applied force F .

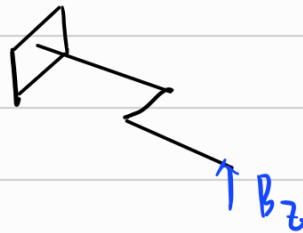


By using superposition method-

System "0"



System "1"



$$I = \frac{\pi r^4}{4} \quad I_T = \frac{\pi r^4}{2}$$

$$E/G = \frac{3}{8} \quad E = \frac{3}{8}G$$

$$G = \frac{8}{3}E$$

$$\frac{E}{G} = \frac{8}{3}$$

$$I_T = 2I$$

$$w'_{max} = \frac{Fl^2}{2EI} \quad w_{max} = \frac{Fl^3}{3EI}$$

$$w_B^{(0)} = \frac{F(2l)^3}{3EI} + \frac{Fl^3}{3EI} + \frac{F(2l)^2}{2EI}(2l) + \frac{(Fl)(2l)}{GI_T}l$$

$$= \frac{8Fl^3}{3EI} + \frac{Fl^3}{3EI} + \frac{4Fl^3}{EI} + \frac{2Fl^3}{GI_T}$$

$$\frac{2Fl^3}{\frac{8}{3}E \times 2I} = \frac{3}{8} \frac{Fl^3}{EI}$$

$$= \left(\frac{8}{3} + \frac{1}{3} + 4 + \frac{3}{8} \right) \frac{Fl^3}{EI} = \frac{59}{8} \frac{Fl^3}{EI}$$

$$w_B^{(1)} = \frac{Bz(4l)^3}{3EI} + \frac{Bz(2l)^3}{3EI} + \frac{Bz(2l)(2l)}{GI_T}l + \frac{Bz(2l)l}{GI_T}2l$$

$$= \frac{64 B_2 l^3}{3 EI} + \frac{1}{3} \frac{B_2 l^3}{EI} + \frac{2 B_2 l^3}{6 EI} + \frac{4 B_2 l^3}{6 EI}$$

$$= \left(\frac{64}{3} + \frac{1}{3} + \frac{2}{3} + \frac{2}{3} \right) \frac{B_2 l^3}{EI} = \frac{547}{24} \frac{B_2 l^3}{EI}$$

$$\omega_B^{(0)} = \omega_B^{(1)} \quad \frac{59 Fl^3}{8 EI} = \frac{547}{24} \frac{B_2 l^3}{EI} \quad B_2 = \frac{177}{547} F$$

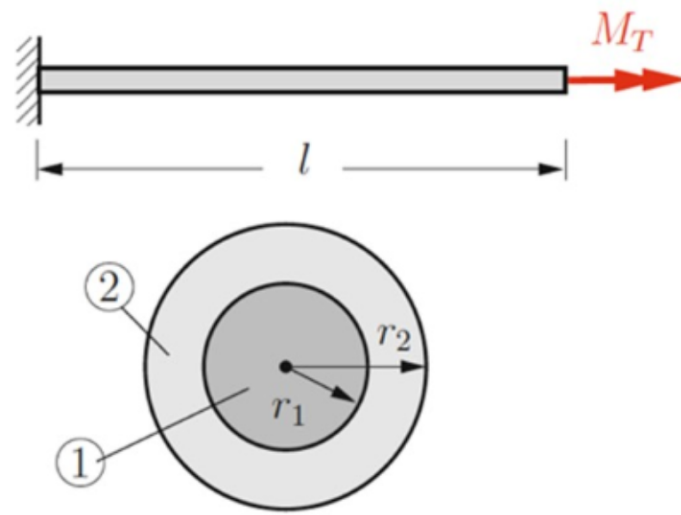
$$A_z + B_z = F \quad A_z = F - B_z = \frac{370}{547} F$$

$$M_{AX} = M_{AX}^{(0)} + M_{AX}^{(1)} = M_{AX}^{(0)} = \frac{370}{547} Fl$$

$$M_{Ay} = \frac{370}{547} F(2l) - \frac{177}{547} F(2l) = \frac{386}{547} Fl$$

Homework 5.2 The shaft shown in the left figure consists of a tube (shear modulus G_2) which is bonded to a core (shear modulus G_1). It is subjected to a torque M_T at its free end.

Determine the maximum shear stresses in the core (1) and in the tube (2) and the angle of twist.



$$\theta_l = \frac{M_T l}{G I_T} \quad \text{since its circular } I_T = I_p$$

$$\theta_{(1)} = \frac{M_{T1} l}{G_1 I_{T1}} \quad \theta_{(2)} = \frac{M_{T2} l}{G_2 I_{T2}}$$

$$\tau_{\max(1)} = \frac{M_{T1}}{W_{T1}} \quad \tau_{\max(2)} = \frac{M_{T2}}{W_{T2}}$$

$$I_{T1} = \frac{\pi r_1^4}{2} \quad I_{T2} = \frac{\pi (r_2^4 - r_1^4)}{2}$$

$$W_{T1} = \frac{I_{T1}}{r_{\max}} = \frac{\pi r_1^4}{2 r_1} = \frac{\pi r_1^3}{2}$$

$$W_{T2} = \frac{I_{T2}}{r_{2\max}} = \frac{\pi (r_2^4 - r_1^4)}{2 r_2}$$

$$M_T = M_{T1} + M_{T2}$$

$$\theta = \theta_{(1)} = \theta_{(2)} \quad \frac{M_{T1} l}{G_1 I_{T1}} = \frac{M_{T2} l}{G_2 I_{T2}}$$

$$M_{T1} = \frac{G_1 I_{T1}}{G_2 I_{T2}} M_{T2}$$

$$M_T = \frac{G_1 I_{T1}}{G_2 I_{T2}} M_{T2} + M_{T2}$$

$$M_T = \frac{G_2 I_{T2} + G_1 I_{T1}}{G_2 I_{T2}} M_{T2} \quad M_{T1} = \frac{G_1 I_{T1}}{G_1 I_{T1} + G_2 I_{T2}} M_T$$

$$M_{T2} = \frac{G_2 I_{T2}}{G_1 I_{T1} + G_2 I_{T2}} M_T$$

$$T_{\max(1)} = \frac{G_1 I_1}{G_1 I_1 + G_2 I_2} M_T \times \frac{r_1}{I_1} = \frac{G_1 r_1}{G_1 I_1 + G_2 I_2} M_T$$

$$T_{\max(2)} = \frac{G_2 r_2 M_T}{G_1 I_1 + G_2 I_2}$$

$$\theta = \frac{M_T l}{G_1 I_1 + G_2 I_2}$$